

Modeling Business Processes (Part 2)



Business process modeling is primarily used to map a workflow in order to comprehend, analyze, and improve that workflow or process. The use of a diagram aids in visualizing the process and making better judgments.

Different techniques/ diagrams of BPM:

Let's start with the latest:

1. Business Process Modeling Notation (BPMN)

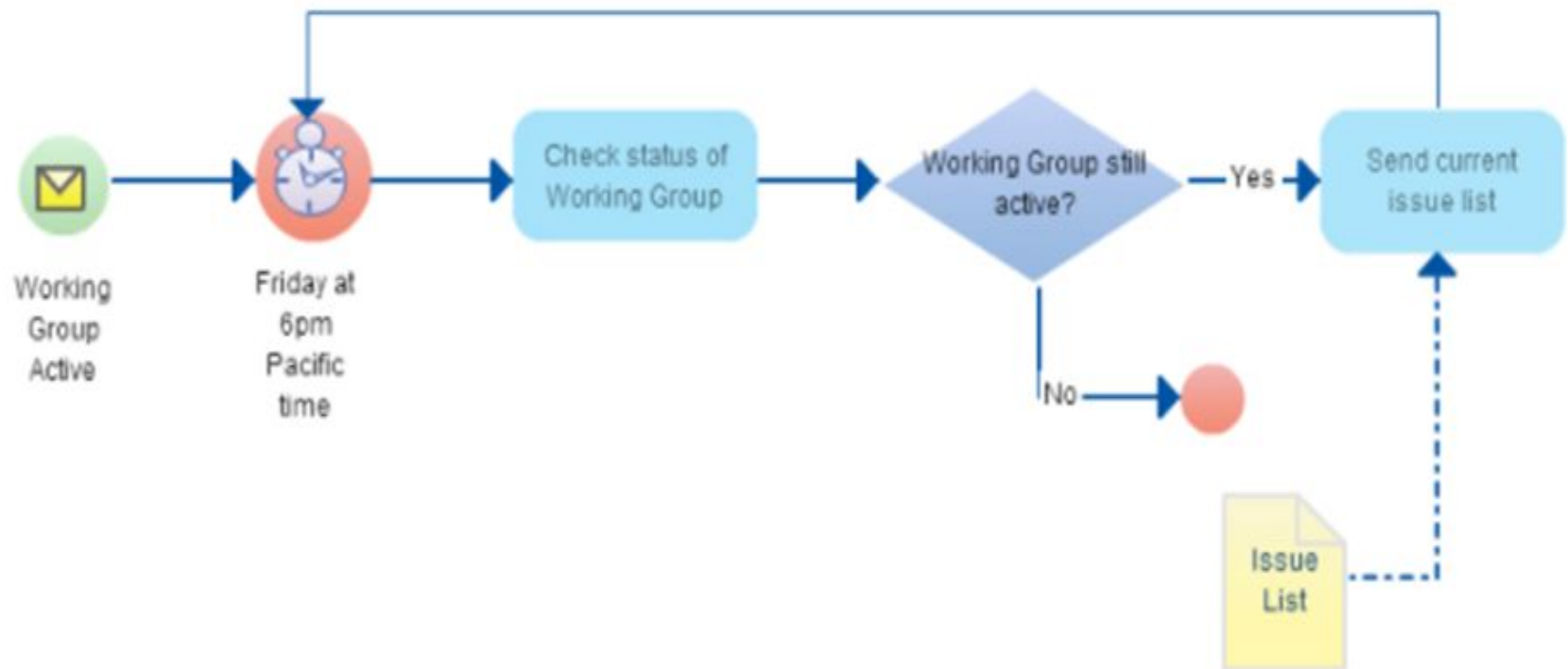
- Simply defined, BPMN is a standard object-based graphical representation of your business process. It can also be characterized as a set of graphical objects and rules specifying potential connections between the objects if you want to go more technical.

- **BPMN stands for business process modeling notation, and it is a set of symbols that are used to represent activities and workflows. Any symbols can be used in your business process, but adopting standardized ones makes it easier to communicate with outside analysts and avoids the need to create your own visual language.**

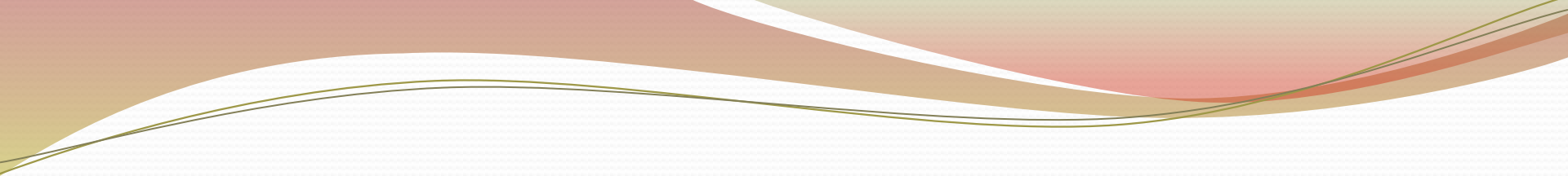
BPMN consists of the following basic building blocks:

- ✓ **Flow objects: events (circles), activities (rectangles with rounded corners), and gateways (diamonds)**
- ✓ **Connecting objects: mainly comprising arrows, these indicate sequence flow (filled arrows), message flow (dashed arrows), and associations**

- ✓ Swim lanes: pools (graphic container) and lanes (sub-partition of the pool)
- ✓ Artifacts: data objects, groups, and annotations

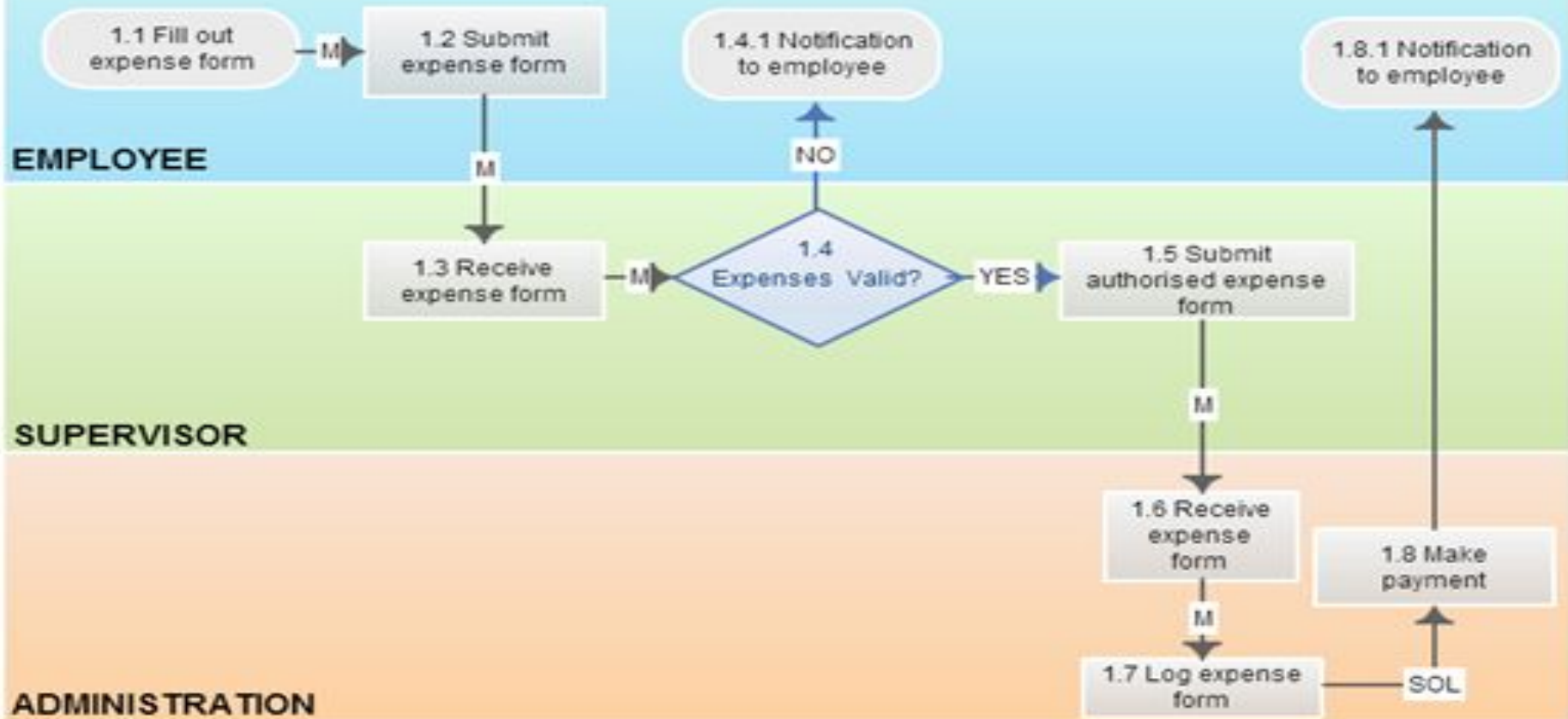


A business process modeled using BPMN



The most significant benefit of BPMN is that it is a well-defined standard. Because so many business analysts are familiar with it, collaboration becomes easier. Furthermore, most modeling tools accept BPMN, making it much easier to share and update even between different programs. All of these factors combine to make BPMN the most widely used business process modeling technique today.

Expense Claim Process



M - Manual
Sol - Facilitated by Solution

A process modeled using BPMN that has swimlanes

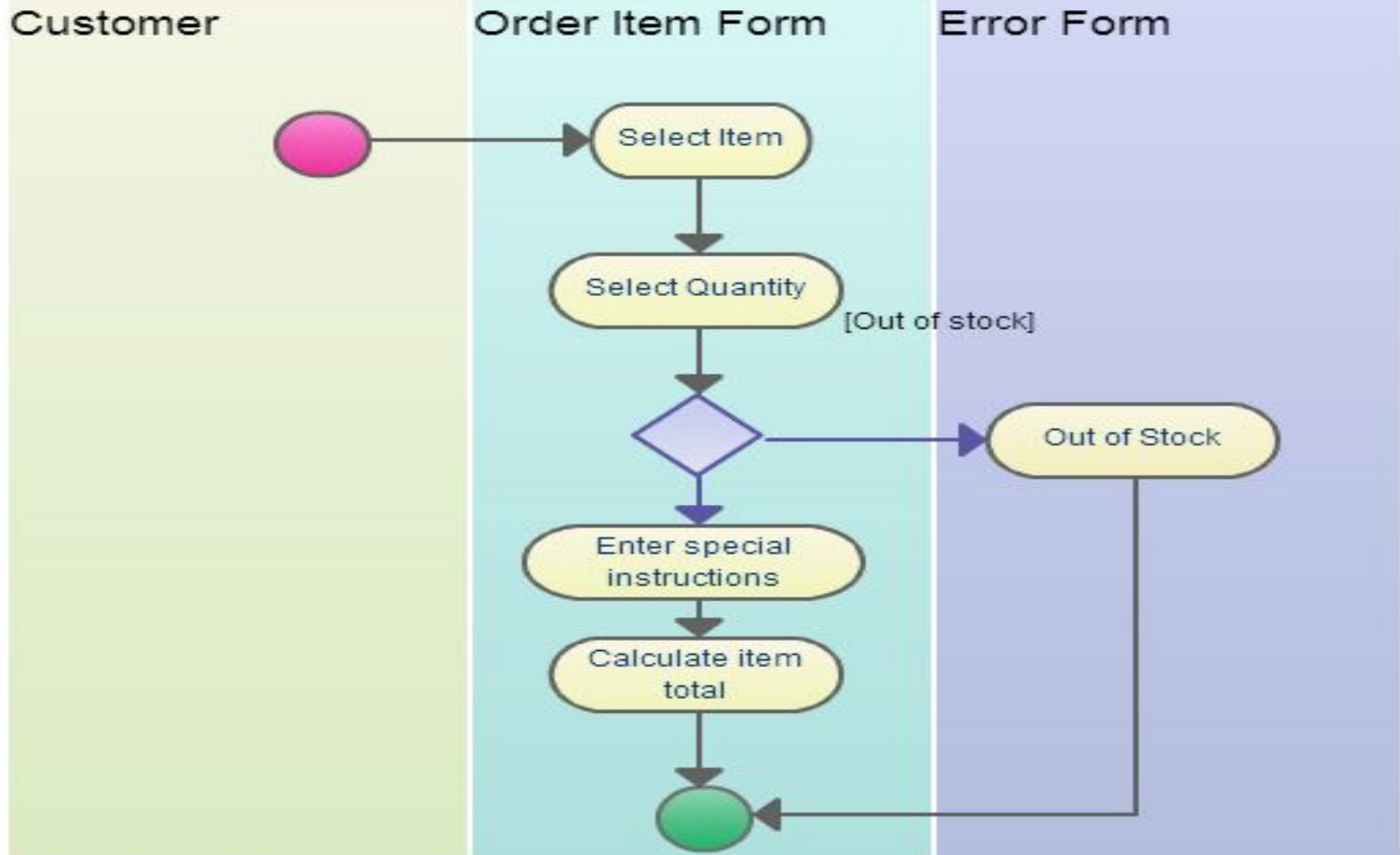
2. UML Diagrams

UML is a modeling language that is primarily used for software system specification, visualization, development, and documentation. However, it has been used by business people as an effective business process modeling tool.

A more current way to modeling and documenting processes is the Unified Modeling Language (UML). UML was created by software professionals and has since been effectively applied to business process modeling.

With 14 different UML diagram types, it gives a versatile and powerful approach to illustrating virtually any business process. They're frequently used to model the detailed logic of a business process. In many ways, UML diagrams are the object-oriented equivalent of flow charts.

One of its key advantages, as previously stated, is its versatility. However, with 14 various diagram kinds; some people may have trouble understanding the diagrams. Furthermore, numerous UML diagrams can be used to model the same process. As a result, it's unlikely to be the most popular option among analysts.

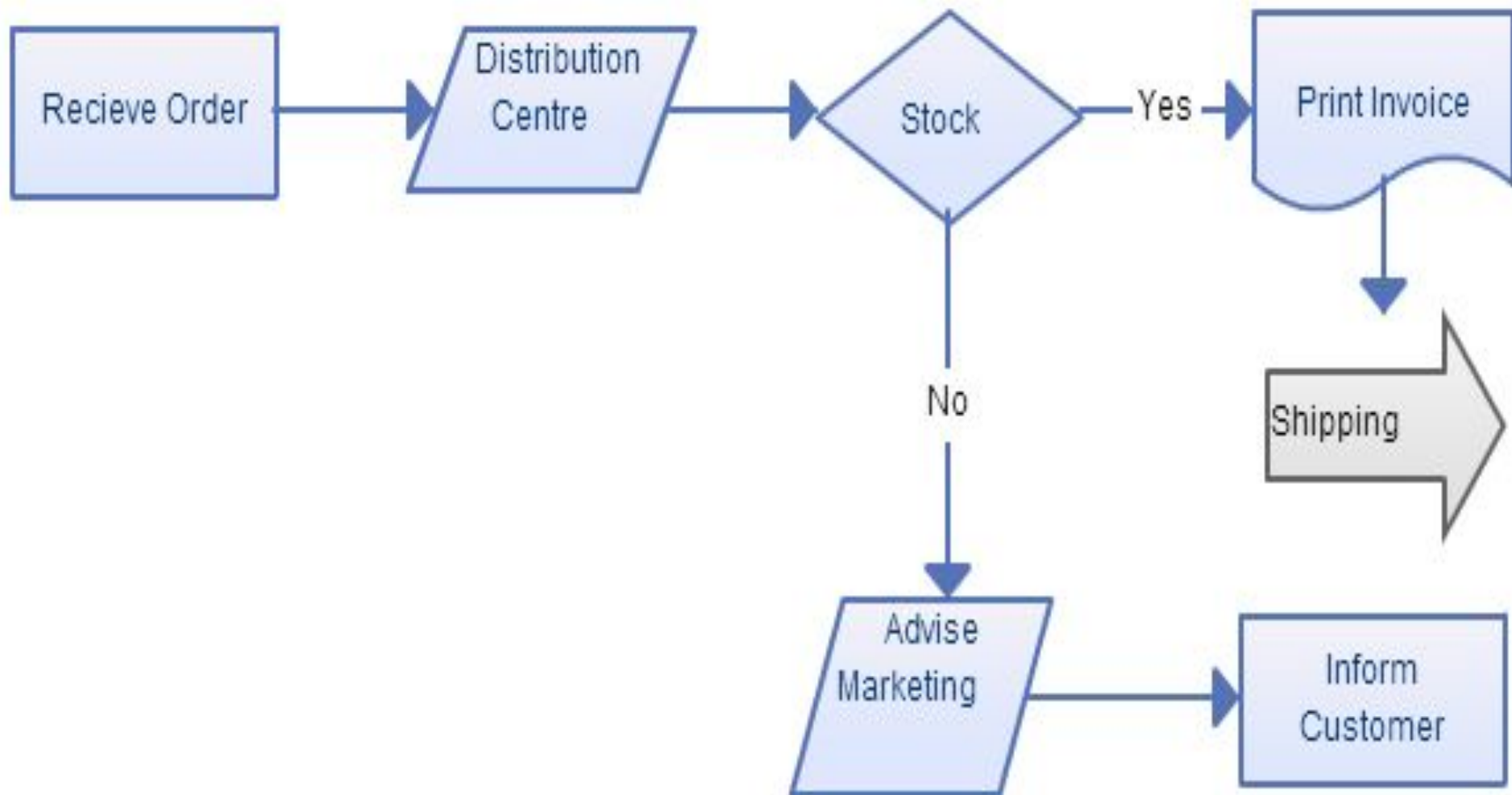


It's flexibility makes UML a popular business process modeling technique

3. *Flowchart Technique*

Flowcharts are the most widely used diagram kind on the planet. Many people can understand it because it only contains a few standard symbols. Because of its simplicity, it is a potent and effective instrument. BPMN is, in fact, a more complex version of the basic flowchart technique. Moreover, most drawing software supports the production of flowcharts; therefore they are used by a much larger audience.

The flowchart employs a sequential sequence of actions and does not allow for activity breakdown. It's possible that the Flow Chart model was the first process notation. Although there is no definite date for its origin, it has been widely used for many years.

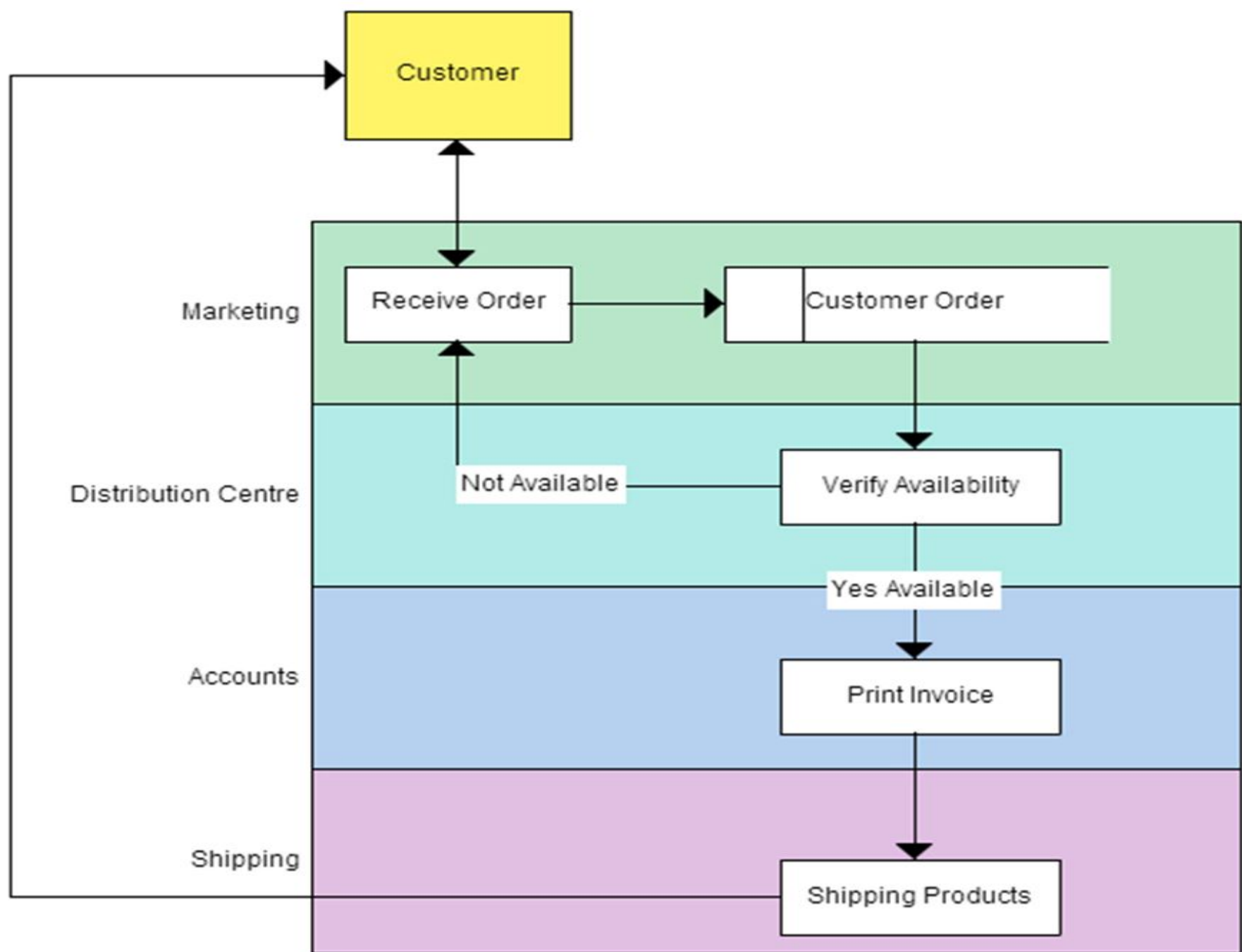


You can quickly create a professional flowchart with our flowchart software.

4. *Data Flow Diagrams – Yourdon's Technique*

Data flow diagrams (DFD) demonstrate the movement of data or information from one location to another. DFDs are diagrams that depict how processes are linked together through data repositories and how they interact with users and the outside world.

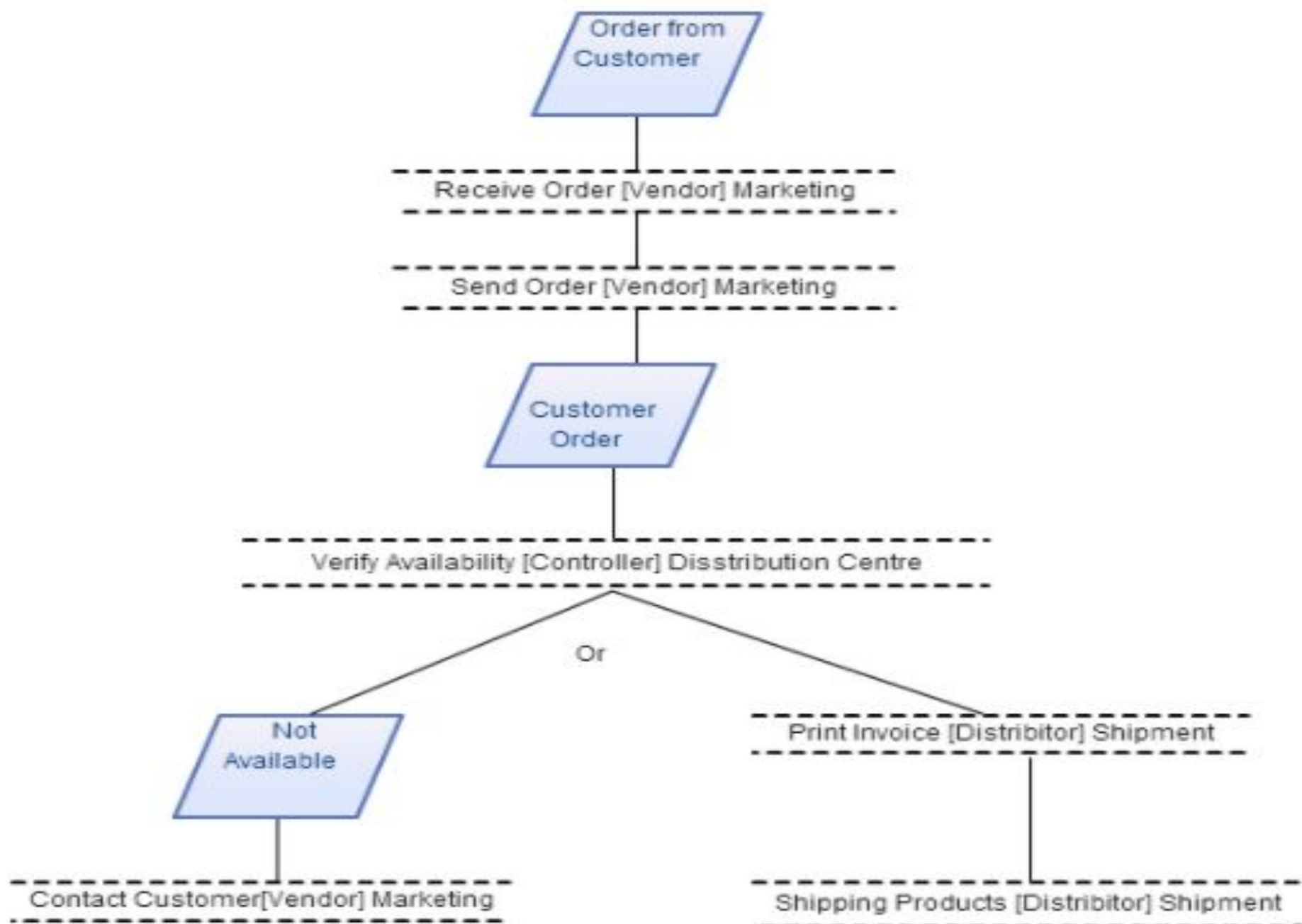
They're utilized to keep track of the processes that are being investigated as part of the design documentation. A DFD can be seen as a way of arranging data in its raw form. DFDs are the foundation of structural analysis, which was created by Yourdon in the early 1960s.



*A DFD diagram used in modeling done by **Data Flow Diagram Tool***

5. *RAD (Role Activity Diagrams)*

Roles are abstract behavioral notations that describe a desired behavior inside an organization. They're frequently administrative functions. Software systems, customers, and suppliers are also included. RADs offer a unique viewpoint on the process and are especially helpful in facilitating dialogue. They are easy to read and comprehend, and they provide a full view of the process and permitting operations at the same time.

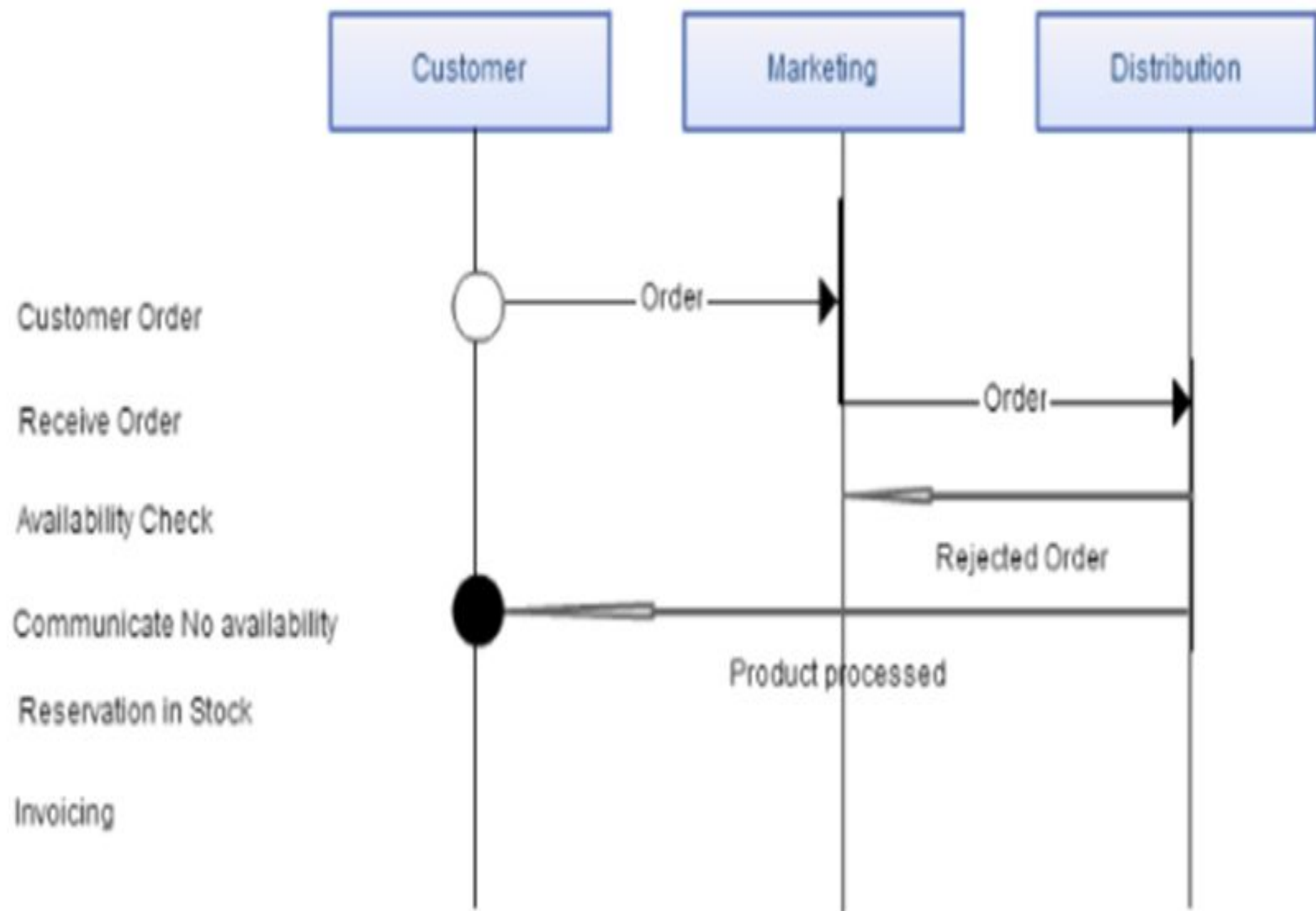


6. *Role Interaction Diagrams – RID*

In a matrix, activities are tied to roles. On the left axis, activities are shown vertically, and roles are shown horizontally at the top. The procedure is represented using a combination of text and symbols.

RIDs are significantly more complex than flow diagrams, but they are rather intuitive to grasp and read. However, they are messy, with numerous arrows pointing left and right, and hence difficult to construct.

The activities' inputs and outputs are not modeled. As a result, critical information is lost. RIDs, for example, are not as adaptable as flowcharts. They have a very strict notation. RIDs, on the other hand, are versatile in comparison to other modeling methodologies.



Role interaction diagrams offer flexibility but somewhat harder to draw

7. Gantt Charts

A list of activities is linked to a time scale using Gantt charts. Its strength rests in its capacity to monitor the current status, project timeline, and resource allocation, despite the fact that it can be used to graphically illustrate a process. Although it can show topics like interdependence, the modeling section is extremely inadequate. This is most likely why it isn't as popular as it once was.

A Gantt chart is a matrix where lists of all tasks and activities to be done in the process are on the vertical axis. Each row has single activity identification, which is commonly a number and a name. Columns indicate estimated activity length, skill level required to do the activity, and the name of the person allocated to the activity on the horizontal axis, which is followed by one column for each phase of the project's lifespan. Hours, days, weeks, months, and other time units can be used to express each period. It may be required to designate the period columns as period 1, period 2, and so on in some circumstances.

Project Plan Overview



Gantt charts are used for time lines in processes

THANK YOU FOR LISTENING



ANY QUESTIONS?